OrderFit

A financial agent that tells a small business whether it can safely accept a new order.

The problem

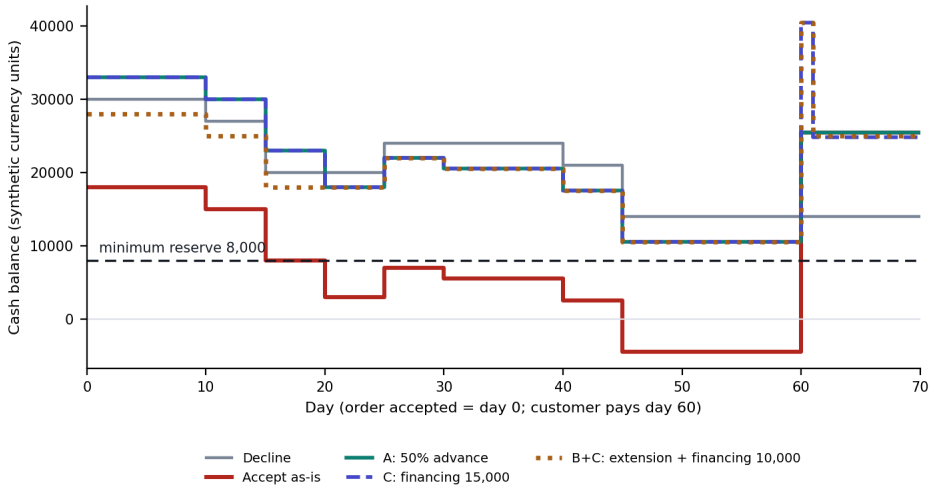
A small business gets a big order. It looks profitable, but the business must pay suppliers and payroll **weeks before the customer pays**. Saying yes can leave it short of cash.

What OrderFit does

1. Reads the order and the supplier quote.
2. Maps day by day when cash goes out and comes in.
3. Checks that cash stays above the owner's minimum reserve.
4. If it does not, compares the fixes: a customer advance, a later supplier payment, a financing offer, or declining.
5. Recommends one, drafts the message, and after the owner approves, carries it out in a **simulated** bank.

Example (invented numbers): a 30,000 order, reserve 8,000

Option	Lowest cash	Meets reserve	Financing cost	If customer pays 15 days late
Accept as-is	-4,500	No	0	Fails
A: 50% advance	10,500	Yes	0	Still meets
B: supplier extension	500	No	0	Fails
C: financing 15,000	10,500	Yes	675	Fails
B+C: extension + financing 10,000	10,500	Yes	500	Fails



Accepting as-is drops cash below the reserve on day 20. A 50% advance fixes it for free. If the customer pays late, the financing plans fail, so the agent marks the answer **uncertain** and asks the owner.

Why it is not just a forecast

- It decides about **one specific order** and compares actions, not only scenarios.
- A fixed calculator does all the math. The AI only ranks options and flags what is missing or uncertain; it cannot approve a plan that breaks a hard limit.
- It completes the task: extract, simulate, decide, draft, then execute in the sandbox with a full log.

How it is built

Part	Job
Extraction (AI)	Pulls dates and amounts from documents, each linked to its source
Financial engine (fixed code)	Exact arithmetic, cash timeline, checks hard limits
Decision model	Ranks feasible options, lists missing info, escalates when unsure
Permission gate	Four tiers: read, propose, act in sandbox after approval, never (live bank or real lender)
Sandbox executor	Simulated bank, mock email outbox, mock lender portal, operation log

Fit with the rules (Topic E)

- **Covers:** financing-option matching, cash-flow management and alerts, business financial analysis.
- **Scoring:** task completion 40% (end-to-end run and sandbox log), security 30% (permission tiers, injection tests), innovation 30% (order-level decision, explanation).
- **Submit by 20 Oct, 23:59 Beijing time:** tech document, deck, optional video, code with tests, security self-assessment, sandbox logs.

How we prove it works

- 10 test scenarios with known answers, checked against a separate reference calculation.
- Compared with three baselines: profit-only, fixed rule, and an AI reading the documents alone.
- Measured: calculation accuracy, rule violations (target 0), decision quality, correct escalation, injection resistance.

Main risks and questions for the mentor

- We have not shown owners need this, or that Xero-type tools do not already do it.
- Is deciding on one order really different from cash-flow forecasting?
- Does the sandbox log count as task completion for this topic?